Nur Mohammad

nmohammadnurbd@gmail.com | ResearchGate | Google Scholar

Research Interests

Human-Environment Interactions, Remote Sensing, GIScience, Environmental & Vegetation Change Monitoring, and Climate Change

Education

MSc in Cartography *Sep'23 – Jul'25*
Eötvös Loránd University, Budapest, Hungary
CGPA: 4.94/5.00
Thesis: 'Mapping of Daily Light Integral (DLI) Values in Bangladesh with Alternative Scaling Approaches.'

MSc in Geography and Environment *Dec'21 – Feb'23*
Jahangirnagar University, Dhaka, Bangladesh
CGPA: 3.93/4.00
Thesis: 'Mapping Historical Changes and Village Impacts along the Dhaleswari River Using Integrated Geospatial Data.'

BSc in Geography and Environment *Mar'16 – Feb'21*
Jahangirnagar University, Dhaka, Bangladesh
CGPA: 3.84/4.00
Dissertation: 'Monitoring SDG's Indicator for Urban Built-up Areas Expansion: A Remote Sensing Based Study on Dhaka City Corporation.'

Professional Experience

PlantVision, Hungary
Geospatial Research Assistant *Sep'25 – present*

- Develop automated Python workflows for spectral and photoreceptor data processing, calibrate sensors using ENVI and ArcGIS, and author technical reports documenting vegetation monitoring in Hungary.

Centre for Environment and Geographic Information Services (CEGIS), Bangladesh

1. **Research Consultant** *Feb'23 – Jul'23*
   - Delivered plot-level LULC geodatabases using high-resolution imagery and RTK-GNSS ground control points, supporting national resource planning with sub-meter georeferencing accuracy.
2. **Technical Research Consultant** *Feb'21 – Dec'21*
   - Architected GIS geodatabases from legacy cadastral maps, maintained satellite-derived land-use databases, and directed quality control workflows to meet government standards for national census planning.

Living Lab for Rural Studies, Jahangirnagar University, Bangladesh
Research Assistant *Jan'18 – Dec'21*

- Executed integrated GPS and satellite imagery surveys in ArcGIS, analyzed physical and cultural features via Google Earth Pro, and led LULC classification and DEM-based analysis contributing to peer-reviewed.

River and Delta Research Center (RDRC), Bangladesh
Research Assistant *Jan'20 – Jan'21*

- Led satellite imagery and GPS-based river boundary analysis, supported field methodology development, and contributed to data analysis, report writing, and peer-reviewed publishing.

Academic Care Center (ACC), Bangladesh
Geography Tutor *Mar'16 – Dec'20*

- Designed and delivered geography curriculum covering geomorphology, climate change, and cartography, and assessed student progress through structured tests and quizzes.

Skills

Geospatial Methods	Remote sensing, GIS-based spatial analysis, UAV-based data acquisition, LULC classification, and time-series analysis
Machine Learning	Machine learning; deep learning (CNN-based classification)
Programming	Python, JavaScript, HTML, CSS, PostgreSQL
Geospatial Tools	ArcGIS Pro, QGIS, Earth Engine, ERDAS Imagine, SNAP, Global Mapper

Scholarly Publications

Peer-reviewed Articles

1. Habib, M. A., Kabir, F., & **Mohammad, N.** (2024). Pixelizing and Validating Land Surface Temperature from Band 10 of Landsat 8. *The Jahangirnagar Review: Part II: Social Science*, 48(2), 77–96. DOI
2. Hasan, M. R., **Mohammad, N.**, Tahsin, M., & Mollah, T. H. (2023). Predicting spatial-temporal assessment of threatened wetland and sensitive ecosystem for 2029 using CA-Markov chain model, Bangladesh. *Journal of Life and Earth Sciences*, 1, 65–81. DOI
3. Habib, M. A., Kabir, F., **Mohammad, N.**, & Rashid, M. S. (2022). Estimating SDG Targets and Indicators Using GIS and Remote Sensing Techniques: A Study in Savar Upazila. *Jahangirnagar bishwavidyalay bhugol O paribesh samikkhan*, 41, 11–24. (in Bengali) DOI
4. Hossain, M. J., **Mohammad, N.**, Jahan, N., Tabassum, M., & Zoha, K. R. (2022). Examine the Existing Solid Waste Management Practices for Developing a ISWM's Framework at Jahangirnagar University. *Research and Reviews: Journal of Environmental Sciences*, 4(2), 1–21. DOI
5. Mollah, T. H., Hasan, M. R., **Mohammad, N.**, & Tahsin, M. (2021). Land Use/Land Cover (LULC) Changes Mapping: A Case on St. Martins Island, Bangladesh. *Somajbiggan somikkha*, 5, 49–58. (in Bengali) DOI
6. Mollah, T. H., Tahsin, M., **Mohammad, N.**, Mollah, N., Hasan, M. R., & Nihei, T. (2021). Exploring Role of Local NGOs in Agriculture: a GIS Based study along the Jamuna River, Bangladesh. *The Jahangirnagar Review: Part II: Social Science*, XLV, 151–170. DOI
7. Mollah, T. H., Tahsin, M., **Mohammad, N.**, Hasan, M. R., & Mollah, N. (2021). Predicating Land Use/Land Cover Changes for 2050 Using CA-Markov Model and LCM: A Case for Maheshkhali Island, Bangladesh. *The Jahangirnagar Review: Part II: Social Sciences*, XLIV, 213–225. DOI
8. Mollah, T. H., **Mohammad, N.**, Tahsin, M., Hasan, M. R., & Nihei, T. (2019). SWOT Analysis for Local NGOs: A Geo-spatial Analysis Evidence from Rural Bangladesh. *The Jahangirnagar Review: Part II: Social Sciences*, XLIII, 179–194. DOI

Conference Abstracts/Short Papers

9. **Mohammad, N.** & Jahan, N. (2026). Mapping Urban Cooling Inefficiency Hotspots in the Dhaka Region Using Deep Learning and Landsat-Derived Indicators. *AGIT Conference: Shaping Geospatial Futures*, Vol. 2(1), pp. 163–164. DOI
10. **Mohammad, N.**, Jahan, N., Hossain, M. J., Jung, A., & Rashid, M. S. (2025). Spatiotemporal Correlation Between Biophysical Parameters and Land Surface Dynamics in an Archaeological Region of Bangladesh. *AGIT Conference: Shaping Geospatial Futures*, Vol. 1(1), 138–145. DOI
11. **Mohammad, N.**, Jahan, N., & Rashid, M. S. (2022). Monitoring SDG's Indicator for Urban Built-up Areas Expansion: A Remote Sensing Based Study on Dhaka City Corporation. *International Conference on Environmental Protection for Sustainable Development (ICEPSD)*, Dhaka, Bangladesh. DOI

Book Chapter

12. Rashid, M. S., Jahan, N., & **Mohammad, N.** (2025). Untapping the Potentials of Mauza Maps in Detecting Coastal Erosional Hazard of Bangladesh by Monitoring Coastline Shifting: A GIS and Remote Sensing Based Analysis. *Contemporary Disaster Issues in Bangladesh*, University of Dhaka.

Submitted Articles

13. **Mohammad, N.**, Jahan, N., Varga, Z., Jung, A., & Sipos, L. (2026). Daily Light Integral Mapping of Bangladesh: Monthly, Seasonal, and Physiographic Patterns. *Discover Sustainability*.(Under review)

Conference Presentations

1. **Oral Presenter**, *AGIT 2026 Conference for Geoinformatics: Shaping Geospatial Futures*, Salzburg, Austria; 8–9 Jul 2026.
2. **Oral Presenter**, *AGIT 2025 Conference for Geoinformatics: Shaping Geospatial Futures*, Salzburg, Austria; 2–3 Jul 2025.
3. **Poster Presenter**, *International Conference on Environmental Protection for Sustainable Development (ICEPSD)*, Dhaka, Bangladesh; 2–4 Sep 2022.

Awards and Honors

1. **Excellent Student Research Scholarship**, Faculty of Informatics, Eötvös Loránd University, 2024
2. **Stipendium Hungaricum & Diaspora Scholarship Excellence Award**, Tempus Public Foundation, 2024
3. **Stipendium Hungaricum Scholarship**, Eötvös Loránd University and Tempus Public Foundation, 2023
4. **National Science and Technology (NST) Fellowship**, Ministry of Science and Technology, 2021
5. **Habibur Rahman Kabir Memorial Scholarship**, Jahangirnagar University, 2019
6. **Jahangirnagar University Merit Scholarship**, Jahangirnagar University, 2016-2020

Service & Leadership

Member Coordinator and Volunteer *Feb'16 – Jun'22*
Lighter Youth Foundation, Bangladesh

- Led flood relief operations and youth engagement programs, coordinating a 16-member volunteer team to strengthen community resilience.

Organizing Secretary *Jan'19 – Dec'19*
Jahangirnagar University River Conservation Club, Bangladesh

- Organized seminars, workshops, and awareness campaigns on river conservation, collaborating with faculty and students on environmental field activities.

Volunteer *Jan'18 – Dec'18*
Riverine People Bangladesh

- Conducted field investigations on river pollution and encroachment, contributing to documentation, outreach, and advocacy programs for river sustainability.

Professional Development (Selected)

1. **Python Programming, Data Analysis, Visualization and Statistical Analysis**, Research and Skill Development Society *May–Jul'24*
2. **R Programming**, Jahangirnagar University Higher Study Club *Oct–Nov'20*
3. **IGARSS Online Summer School: Remote Sensing for Volcanos**, IEEE *Aug'20*
4. **Data Analysis using SPSS**, Institute of Information and Technology, Jahangirnagar University *Jan–Apr'19*
5. **Basic GIS and Remote Sensing**, Institute of Remote Sensing and GIS, Jahangirnagar University *Feb'18*
6. **Online Courses**: GIS, Mapping, and Spatial Analysis (Coursera); Google Earth Engine (Udemy); ArcGIS Pro (LinkedIn Learning); QGIS 3 (Geo University); Cartography & GIS for Climate Action (ESRI MOOC)